

Response to Referee Reports

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Response to Referee 1

Point 1 (R1-1): Physical grounding of the surface-diffusion parameter T_s

Referee Comment

T_s controls the number of post-attachment lateral diffusion steps and therefore governs the ratio between the molecular relaxation timescale and the fibril growth rate. In real type I collagen fibrillogenesis, this ratio is modulated by temperature, collagen concentration, pH, and ionic strength, all experimentally accessible quantities. The authors acknowledge that T_s has no direct mapping to biochemical conditions (e.g., pH), yet proceed to speculate in the concluding paragraph about evolutionary fine-tuning of fibril compactness and its role in diseases such as pulmonary emphysema and aortic aneurysm. These biological inferences are unsupported without at least a qualitative discussion of how T_s relates to measurable physical or physiological variables. The manuscript would be strengthened considerably by such a discussion, even at the order-of-magnitude level.

Author Response

We thank the Referee for highlighting the need to clarify the physical basis of T_s . During collagen fibrillogenesis, the resulting molecular organization depends on the competition between the incorporation of new material and the opportunity for local rearrangement during growth. We denote by τ_{dep} the characteristic timescale of molecular incorporation within a local growth region and by τ_{rel} the characteristic timescale of local surface rearrangement. Accordingly, the ratio $\tau_{\text{dep}}/\tau_{\text{rel}}$ provides a conceptual measure of the opportunity for rearrangement before continued growth modifies the local environment. A larger ratio corresponds to more effective relaxation relative to incorporation, whereas a smaller ratio corresponds to growth with more limited relaxation.

Experimental studies show that temperature and collagen concentration influence nucleation and assembly kinetics, as well as the resulting fibril morphology (Kadler et al., 1988; Yang and Kaufman, 2009; Gobeaux et al., 2008). In turn, pH, ionic strength, and buffer composition modify electrostatic interactions, lateral association, and fibril growth kinetics (Kalbitzer and Pompe, 2018; Morozova and Muthukumar, 2018). These conditions may affect incorporation and relaxation simultaneously, thereby changing the effective balance represented by $\tau_{\text{dep}}/\tau_{\text{rel}}$.

Our algorithm represents this physical competition in a discrete, coarse-grained form through T_s . After attachment, each molecule is allowed up to T_s lateral surface-diffusion attempts to search for a configuration with lower exposed surface area. Consequently, increasing T_s represents a greater effective opportunity for post-attachment relaxation relative to continued growth. The parameter is not itself a physical time and does not represent a direct change in any single experimental variable. Rather, it captures the net effect that physicochemical assembly conditions may exert on the relative opportunity for local rearrangement during growth.

We have revised the model description to make this interpretation explicit. Because the model contains no calibrated physical time, it cannot presently assign a unique or quantitative value of T_s to a particular temperature, pH, collagen concentration, ionic strength, or buffer composition. Establishing even an order-of-magnitude correspondence would require experimental estimates of the relevant incorporation and rearrangement timescales, followed by calibration against measured growth kinetics or fibril morphology. We have also removed the evolutionary and disease-specific extrapolations from the conclusion, which is now limited to the model result that greater effective post-attachment relaxation produces more compact fibrils and greater resistance to failure in the simulations.

Point 2 (R1-2): Attribution of Self-Organized Criticality

Referee Comment

The authors claim that the rupture process exhibits SOC on the basis that the avalanche size distribution follows a power law. This reasoning is insufficient. SOC requires that a system spontaneously evolves toward a critical state without external tuning of a control parameter, the canonical example being the Bak–Tang–Wiesenfeld sandpile. In the present model, the external force F is continuously increased by the experimenter until failure occurs. The system is driven, not self-organizing.

Critically, two of the papers the authors themselves cite to support the SOC claim (refs. [42] and [43], Zapperi et al., PRL 1997 and PRE 1999) make no reference to SOC. Instead, Zapperi et al. explicitly characterize the breakdown point as a first-order transition and draw an analogy with spinodal nucleation, a fundamentally different phenomenology. The correct terminology for the dynamics observed here is “avalanche statistics in driven disordered fracture,” not SOC.

Furthermore, the power-law regime in Fig. 9(a) spans at most two orders of magnitude in avalanche size, a marginal range. No statistical validation of the power-law hypothesis is presented (e.g., maximum likelihood estimation of the exponent, Kolmogorov–Smirnov test, or comparison with alternative distributions following the methodology of Clauset et al., SIAM Review 51, 661, 2009). The claim of scale-free behavior therefore rests on insufficiently rigorous grounds.

Author Response

We thank the Referee for this important correction. We agree that the observation of an approximately linear regime in a binned log-log avalanche-size distribution is not sufficient to establish Self-Organized Criticality. In our model, the external force F is progressively increased, damage is irreversible, and failed molecules do not recover. The system therefore does not autonomously evolve toward a stationary critical state.

We have removed the SOC and scale-free terminology throughout the manuscript and now describe the observed behavior as stochastic local-avalanche dynamics in a driven disordered fracture process. We have also removed the interpretation of the fitted slope as evidence of a change in criticality, load-sharing universality, or a crossover between local and global load sharing. Until the distributional hypothesis is tested on the raw discrete event sizes, the existing straight-line fits to the binned data are presented only as descriptive summaries and not as statistical validation of a pure power law. The requested maximum-likelihood estimation, goodness-of-fit testing, and comparison with alternative distributions will be addressed separately in our statistical reanalysis.

Point 3 (R1-3): Interpretation of $\gamma > 5/2$ in fiber-bundle theory

Referee Comment

The authors interpret their measured exponent $\gamma \approx 2.8$ (at high T_s) as evidence of a regime beyond equal load sharing (ELS), signaling a crossover to global load-sharing universality. This interpretation is incorrect.

In fiber-bundle models, $\gamma = 5/2$ is the exact mean-field result for ELS, the limiting case of maximally global stress redistribution (Hemmer & Hansen, J. Appl. Mech. 59, 909, 1992; Daniels, Proc. R. Soc. London A 183, 405, 1945). There is no established theoretical framework that predicts or interprets exponents above $5/2$ in this context.

Moreover, Pradhan, Hansen & Chakrabarti (Rev. Mod. Phys. 82, 499, 2010), also cited by the authors, demonstrate that within ELS, the avalanche exponent undergoes a crossover from $\gamma = 5/2$ (generic regime, away from failure) to $\gamma = 3/2$ (near the critical breakdown point). That is, as stress redistribution becomes maximally global near rupture, the exponent DECREASES, not increases. The authors' claim that $\gamma > 5/2$ signals increasingly global load sharing is therefore the opposite of what the established theory predicts.

The observed deviation from $5/2$ is more plausibly attributable to finite-size effects, limited ensemble statistics (10 fibrils per T_s value), the restricted range over which the power law is fitted, or the specific definition of avalanche adopted in the model. A sensitivity analysis with respect to the Weibull modulus m (fixed at $m = 2$ throughout, following Ref. [35]) would also help assess the robustness of the exponents.

Point 4 (R1-4): Cross-sectional fractal dimension and three-dimensional mechanical structure

Referee Comment

The fractal dimension is measured in 2D but used to characterize a 3D mechanical problem. D_f is estimated from 2D cross-sectional projections (x - z plane), yet the backbone identification and the entire fracture simulation are three-dimensional. The relationship between the 2D cross-sectional D_f and the 3D structural properties that govern mechanical response is not established. The authors use the 2D D_f as a proxy for overall fibril compactness without justification. This connection should either be derived or explicitly discussed as an assumption with its limitations.

Author Response

We thank the Referee for pointing out that the original manuscript did not adequately establish the connection between the cross-sectional fractal dimension and the three-dimensional mechanical structure. The cross-sectional D_f characterizes transverse packing, which is directly relevant to the rupture model because the tensile stress at each axial position is determined by the load-bearing area $N(i)$ of the corresponding cross-section, while molecular resistance depends on coordination within the three-dimensional backbone.

To test this connection explicitly, for each T_s we obtained D_f from the ensemble mass–radius curve constructed using 11 transverse cross-sections from each of 50 independent fibrils. From the corresponding three-dimensional load-bearing backbones, we measured four mechanically relevant descriptors: the mean cross-sectional load-bearing area $\langle N \rangle$; the mean molecular coordination $\langle K \rangle$; the axial coefficient of

variation of the load-bearing area,

$$\text{CV}(N) = \frac{1}{\langle N \rangle} \left[\frac{1}{S_y - 1} \sum_{i=1}^{S_y} (N(i) - \langle N \rangle)^2 \right]^{1/2},$$

which quantifies the relative variation of $N(i)$ along the 201 axial layers; and the mean molecular stress exposure at unit force, $\langle \sigma_M \rangle_{F=1}$, obtained by evaluating the molecular stress defined in Eq. (3) at $F = 1$ and averaging it over the backbone molecules. These descriptors were averaged over the independent fibrils, with their uncertainties reported as standard errors of the mean. The uncertainty in D_f corresponds to the standard error of the linear mass–radius fit. Because the relationships are monotonic and approach plateaus rather than remaining linear, associations were evaluated using Spearman’s rank correlation coefficient, ρ (Spearman, 1904; Schober et al., 2018), between the ten T_s -averaged conditions.

The cross-sectional D_f was positively associated with the mean load-bearing area ($\rho = 0.988$) and mean molecular coordination ($\rho = 1.000$), and inversely associated with the axial variability of the load-bearing area ($\rho = -0.782$) and mean molecular stress exposure at unit force ($\rho = -0.964$). Thus, increasing D_f accompanies cross-sections that contain more load-bearing material, greater molecular coordination, reduced axial heterogeneity, and lower mean stress exposure for the same applied force.

Taken together, these results provide direct numerical validation that D_f captures structural changes relevant to axial load bearing: increasing D_f corresponds to larger, better-coordinated, and more axially uniform load-bearing backbones, with lower mean stress exposure for the same applied force. Within its stated scope, which is the comparison of fibril ensembles generated at different T_s , D_f therefore provides a compact and quantitatively supported proxy for mechanically relevant transverse compactness. We have revised the manuscript to make this interpretation and its scope explicit.

Point 5 (R1-5): Empirical structural-mechanical connection

Referee Comment

The key result of the paper, that D_f and γ saturate together at $T_s \geq 512$, is presented as if it were a derived causal relationship. In fact, it is a numerical correlation observed in simulation. No theoretical framework is provided that connects the fractal dimension of the fibril cross-section to the avalanche exponent of the fracture process. The statement that “the fibril’s fractal dimension serves as a quantitative bridge between structural compactness and failure statistics” in the Conclusion overstates what has been demonstrated. The authors should reframe this as an empirical observation and discuss what theoretical framework might, in the future, explain such a connection.

Point 6 (R1-6): Molecular aspect ratio

Referee Comment

The aspect ratio of model molecules (18:1) differs substantially from that of real collagen molecules ($\approx 200:1$). The impact of this simplification on the fractal dimension and packing geometry should be acknowledged.

Author Response

We thank the Referee and agree that the reduced molecular aspect ratio is an important limitation. Our model adopts the $18 \times 1 \times 1$ lattice-rod representation used by Parkinson et al. (1997), corresponding to an aspect ratio of 18:1. This provides continuity with the original collagen aggregation and fracture framework, but remains a substantial geometric simplification relative to the approximately 200:1 aspect ratio of real collagen molecules.

The reduced aspect ratio may affect molecular overlaps and local coordination, packing and void geometry, and consequently the measured cross-sectional fractal dimension D_f and the mechanically relevant quantities $N(i)$, $CV(N)$, and K . Since $N(i)$ and K determine molecular stress exposure and removal probability in the rupture model, quantitative damage curves, rupture resistance, and avalanche statistics may also depend on the assumed aspect ratio.

Although the same aspect ratio is used for all T_s , providing a consistent geometry for comparisons within the model, this does not establish that the numerical values or observed trends are independent of the aspect-ratio approximation. Because an aspect-ratio sensitivity analysis was not performed, the results should be interpreted as trends within the present coarse-grained model rather than as quantitative predictions for real collagen fibrils. We have revised the Methods to cite the origin of the 18:1 representation and expanded the limitations section to state its possible structural and mechanical consequences.

Point 7 (R1-7): Phenomenological damage function

Referee Comment

The phenomenological function $f(F)$ in Eq. (5) is purely empirical. Its form is not derived from the model and the physical interpretation of the parameters α and β , while discussed qualitatively, would benefit from more precise justification.

Response to Referee 2

Point 1 (R2-1): Loading protocol and stability of the system

Referee Comment

My main concern is the loading protocol. Equation (4) defines the probability of failure. As long as fibers carry load, this probability remains finite. This seems to imply that, at any finite load, the bundle would fail after a finite time, with a characteristic time depending on the load level. The authors state that during a sweep through the system, each fiber is given a chance to fail according to Eq. (4), and if no failure occurs, the external load is increased. This procedure appears somewhat artificial. If I understand the model correctly, the bundle is not fully relaxed after a sweep: failure could continue in a subsequent sweep at the same load level. Moreover, fiber removal increases (σ_M) and decreases (K), both of which increase the rupture probability (P_R). Thus, the system does not seem to be in a stable state when the external load is further increased. The authors should clarify this point in the manuscript.

Author Response

We thank the Referee for pointing out that the meaning of a no-removal sweep was insufficiently explained. Our rupture model does not introduce physical time, a dwell time at fixed load, or a thermally activated failure rate. Accordingly, P_R is a dimensionless removal probability evaluated during a discrete damage-update sweep, not a continuous-time hazard rate.

The loading and stopping rule follows the procedure introduced by Parkinson et al. (1997). At a prescribed force F , every molecule in the load-bearing backbone is assessed once. If removals occur, the backbone is reassessed and another damage sweep is performed at the same force. The force is increased by $\Delta F = 0.5$ only after a complete sweep produces no new removal event, as specified in the original model.

We agree that a no-removal sweep does not establish an absorbing stochastic equilibrium. Because P_R can remain nonzero, another independent assessment of the same unchanged configuration could produce additional removals. The first no-removal sweep is instead the operational stopping criterion of this discrete, progressively stepped loading protocol, separating the damage response assigned to the current force from that produced after the next force increment.

Consequently, the model characterizes failure under stepped loading and is not intended to describe creep, fatigue, delayed rupture, or lifetime under a force held constant for a physical duration. We have revised the Methods to make this interpretation and its limitations explicit.

Point 2 (R2-2): Localized versus equal load sharing

Referee Comment

The authors discuss localized and more equal load-sharing regimes in their interpretation. However, in the simulations, load sharing appears to be determined by Eqs. (2) and (3), which seem to impose equal load sharing within a cross section. If I understand correctly, information about the local structure enters only through K in the rupture probability. The authors should discuss more clearly in what precise sense the model realizes localized or equal-load-sharing behavior.

Author Response

We thank the Referee for identifying this ambiguity. We agree that the original interpretation invited the conclusion that our model realizes a crossover between localized and equal load sharing. In particular, we

compared the fitted values of γ with exponents from fiber-bundle theory and described their variation with T_s as evidence of increasingly global load sharing. This interpretation was not supported by the stress-redistribution rule implemented in the model. Numerical proximity between a fitted slope and a theoretical fiber-bundle exponent is not sufficient to establish a common load-sharing mechanism or universality class.

Equation (2) distributes the applied force uniformly among all load-bearing molecular segments within each cross-section. When a molecule is removed, $N(i)$ decreases only in the cross-sections previously occupied by that molecule, and the corresponding stress increase is shared uniformly among the remaining segments within each affected cross-section. The model does not contain a distance-dependent elastic kernel that preferentially transfers this additional stress to the spatial neighbors of the removed molecule.

Local structure enters the rupture probability through the coordination number K . Removing a molecule reduces K for its nearest neighbors and therefore lowers their resistance to subsequent removal. This produces spatially heterogeneous rupture probabilities, but it does not constitute local load sharing in the conventional fiber-bundle sense. Conversely, because $N(i)$ varies along the fibril and molecules span different sets of cross-sections, the model is not a conventional equal-load-sharing fiber bundle governed by a single globally shared stress.

We therefore describe the model as combining uniform load sharing within each cross-section with local coordination-dependent resistance. We have removed the interpretation of γ as evidence of a crossover between local and global load sharing and no longer assign the model to either universality class. The term “local avalanche” now refers only to the spatial connectedness of removed-molecule clusters and does not imply a local-load-sharing mechanism.

Point 3 (R2-3): Definition of avalanches

Referee Comment

Since load sharing according to Eq. (2) is global within a cross section, it seems that no particular stress concentration can develop around failed (removed) fibers in that cross section. If this interpretation is correct, then the definition of avalanches may require further justification. If the stress field is not localized around failed clusters, it is not obvious why avalanches should be defined as steps in the growth of connected failed clusters. An alternative, and possibly more natural, definition would be to regard an avalanche as the set of fibers that fail between two consecutive increments of the external force. The authors should make this aspect of the work clearer.

Author Response

We thank the Referee for this important observation. We agree that Eq. (2) distributes the load uniformly within each cross-section and that our model does not include a localized elastic stress-redistribution kernel. Nevertheless, removing a molecule directly modifies the cross-sectional stresses only in the sections it previously occupied and reduces the coordination number K of its nearest neighbors. Consequently, damage evolution retains a local structural component even though load is uniformly shared within each affected cross-section.

We quantify this component by partitioning all molecules removed at a fixed force F into nearest-neighbor connected clusters. Following the global-avalanche/local-cluster distinction used by [Laurson, Santucci, and Zapperi, Phys. Rev. E **81**, 046116 \(2010\)](#), we refer to each connected component as a local avalanche (or avalanche cluster). Spatially disconnected clusters are counted as distinct events, so multiple local avalanches may occur at the same F . This is an operational classification of local damage and is not intended to imply a standard local-load-sharing mechanism. We have clarified this definition and its physical motivation in the revised Methods.

Point 4 (R2-4): Power-law statistics and SOC

Referee Comment

The interpretation of power-law statistics in terms of self-organized criticality should be treated with caution. In rupture processes the system is gradually destroyed, and there is no healing mechanism by which failed fibers could recover and carry load again. Therefore, the analogy with self-organized critical systems may be problematic unless it is carefully qualified. I suggest that the authors either provide a more detailed justification for this interpretation or reformulate the discussion in more cautious terms.

Author Response

We agree with the Referee. Because the fracture process is externally driven and irreversible, with no healing mechanism, we no longer interpret the avalanche statistics as evidence of Self-Organized Criticality. The revised manuscript instead describes the process as stochastic local-avalanche dynamics during progressive damage in a driven disordered system. We have removed the SOC and scale-free terminology and explicitly state that the current binned linear fits do not by themselves validate a power-law distribution.